CISCO SECURE

HMF Design Clinic Workbook

Cedarline Health Group, a Hybrid Mesh Firewall design mission

Today your team is the Cisco Solution Engineering team walking into Cedarline. You will sit with four real conversations, hear what is keeping them up at night, and answer as Solution Engineers do, with the right solution, why it fits, and the outcome it delivers, all brought together into one Hybrid Mesh Firewall design.

The scenario. Cedarline Health Group is a regional healthcare payer-provider with roughly 8,000 employees across four hospitals and thirty-one clinics, plus a claims-processing business that handles the money. Growth by acquisition (including a second data centre absorbed as-is) and a mid-flight, disorderly move to public cloud have left a fragmented security posture. After a security incident in March, the board asked what would have stopped it, and leadership still cannot answer. Budget and executive attention are approved but explicitly temporary, and the security team is small.

TEAM	MEMBERS (UP TO 8)	DATE

What you will walk away able to do

1. Map Cedarline's stated concerns to the right Cisco Secure solutions.
2. Explain, in the customer's language, how each solution addresses the concern.
3. Connect the four case studies into one coherent Hybrid Mesh Firewall design.
4. Present the business outcome, not just the feature.

How your answers are assessed

Each pain point is scored across three dimensions: Problem Identifier out of 5, Solution Designer out of 10, and Value Communicator out of 10, for 25 per pain point. Four pain points make a case study score out of 100, and the four case studies make an overall score out of 400.

Problem Identifier OUT OF 5

Correctly identified the customer's pain point and the requirements behind it.

Solution Designer OUT OF 10

Recommended the right Cisco solution, architecture, and approach.

Value Communicator OUT OF 10

Clearly explained how the solution works and the customer outcomes it enables.

The three-part answer. For every customer concern you write three things, one for each dimension you are scored on: **the requirement** behind the concern (identify the problem), a **Product / platform** (design the solution), and **how it helps and the outcome** (communicate the value).

Quick start

In your pair, take one case study. For each pain point: name the requirement, pick a product from the cheat-strip, and write how it helps and the outcome. About five minutes per pain point.

Before you submit, self-check

- Problem: did we name the real requirement behind the concern, not just the surface ask?
- Solution: did we recommend a specific Cisco product and the approach, not only a product name?
- Value: did we explain how it works and the outcome for Cedarline?

What a strong answer looks like

This is an example only, from a different customer conversation. The four case studies you will work are different. Notice how the response identifies the problem, designs the solution, and communicates the value.

Our support team uses a public AI chatbot to draft replies, and staff sometimes paste patient details into it. We cannot see or control what leaves the organisation.

PRODUCT / PLATFORM

Cisco AI Defense

SOLUTION ENGINEER RESPONSE

Identify. The real requirement is preventing sensitive patient data from leaving the organisation through an unsanctioned AI app, with visibility and control over how it is used. **Design.** Cisco AI Defense discovers the AI applications in use and applies runtime guardrails to the prompt and response traffic, so risky data is caught in real time. **Communicate.** This lets the support team keep using AI safely rather than banning it, protects patient data and compliance, and adds no friction to their day-to-day work.

HMF product cheat-strip

The products, platforms and features available across the Hybrid Mesh Firewall. Use it to identify which one fits each pain point; which fits which concern is for you to decide.

Tip: detach this page and keep it beside you while you work.

PLATFORMS AND PRODUCTS

Cisco AI Defense	Secures AI applications and models across build and runtime
Cisco Multicloud Defense	One security policy plane across public clouds
Cisco Secure Firewall (FTD)	Cisco's next-generation firewall, Firewall Threat Defense
Cisco Secure Firewall Management Center (FMC)	The centralised manager for the Secure Firewall estate, handling…
Cisco Security Cloud Control	Cloud-delivered management across the firewall and security estate, for…
Cisco Secure Workload	Application-dependency mapping and micro-segmentation
Cisco Identity Services Engine (ISE)	Network identity and access control
Cisco Hypershield	AI-native, highly distributed security enforcement
Isovalent Cilium	eBPF-based Kubernetes networking and policy, giving consistent…
Isovalent Tetragon	eBPF-based runtime security for workloads, detecting suspicious system…

KEY CAPABILITIES AND FEATURES

Encrypted Visibility Engine (EVE)	A Secure Firewall capability that infers threats from encrypted traffic…
Snort ML	A Secure Firewall capability that adds machine-learning detection…
pxGrid	An ISE capability: a context-sharing fabric so security tools can act…
Adaptive Network Control (ANC)	An ISE capability that quarantines or restricts a device in real time…
Security Group Tags (SGT)	Group-based policy tags (Cisco TrustSec) that enable scalable,…

ARCHITECTURE

Hybrid Mesh Firewall (HMF)	Cisco's approach to consistent security enforcement across firewalls,…

Decode the jargon

EU AI Act, Article 15	EU law classifying AI by risk. A 'Provider' must prove robustness and cybersecurity.
MCP (Model Context Protocol)	How AI agents connect to tools and data, creating a new supply-chain surface to secure.
East-west traffic	Traffic between workloads inside the data centre, as opposed to north-south, which flows in and out.
Lateral movement	An attacker hopping from one compromised system to others on the inside.
Micro-segmentation	Fine-grained isolation of individual workloads so a breach cannot spread.

eBPF	Linux kernel technology used to observe and enforce networking and security with low overhead.
EVE (Encrypted Visibility Engine)	Infers threats from encrypted traffic patterns without decrypting it.
Snort ML	Machine-learning detection inside the Snort engine for unknown and zero-day exploits.
pxGrid	Cisco's context-sharing fabric so tools can act on identity, for example an automatic quarantine via ISE.
Hybrid Mesh Firewall (HMF)	Consistent security enforcement across firewalls, cloud gateways, workloads and Kubernetes, managed centrally.

1 Case Study 1

The conversation. Cedarline is standing up AI across regulated workflows and public cloud faster than security can see it. Help them prove compliance, vet what they adopt, and get visibility and guardrails over AI traffic.

Your task. For each pain point below, name the requirement behind the concern, choose the product from the cheat-strip, then write how it helps and the outcome it delivers.

PAIN POINT 1 Case Study 1

> *Under the EU AI Act, we are classified as a 'Provider' for our internal AI systems. We need to demonstrate robustness and cybersecurity compliance to satisfy Article 15. How does Cisco help us meet these specific requirements?*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 2 Case Study 1

> *We are integrating various open-source models and third-party agents into our internal workflows. How can we ensure these assets aren't introducing vulnerabilities into our environment before they even go live?*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 3 Case Study 1

> *We know AI is showing up across our cloud environments faster than security can track it. We're worried we have models, agents, datasets, and MCP-connected tools running without visibility or consistent controls.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 4 Case Study 1

> *We're running AI workloads in cloud accounts, but we don't have a reliable way to see which models and AI assets are actually in use. We're also concerned about workloads reaching out to third-party AI models without security knowing about it. And even when we do know the traffic path, we need a practical way to inspect and enforce guardrails on prompt and response traffic in the cloud without changing every application.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

2 Case Study 2

The conversation. Workloads sprawl across AWS, Azure and on-prem, each with its own policy model. Contain lateral movement, prove consistent controls to auditors, and give Cedarline one place to define and enforce policy.

Your task. For each pain point below, name the requirement behind the concern, choose the product from the cheat-strip, then write how it helps and the outcome it delivers.

PAIN POINT 1 Case Study 2

> *If a breach occurs in one of our public cloud instances, we are terrified that the attacker will move laterally into our core on-premises systems. How can we contain these threats in a multicloud environment?*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 2 Case Study 2

> *Our auditors require proof that our security controls are applied consistently across all our cloud workloads. Gathering this data from different cloud providers is a manual, time-consuming nightmare.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 3 Case Study 2

> *We have workloads spread across AWS, Azure, and on-premises data centers. Managing security policies individually in each cloud environment is creating massive operational overhead and leading to inconsistent security posture.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 4 Case Study 2

> *We have workloads on-prem, in cloud, and in mixed environments, and policy consistency is becoming difficult. Every environment feels different, and we're worried about drift.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

3 Case Study 3

The conversation. A mixed-vendor perimeter, an inherited rulebase and mostly encrypted traffic leave blind spots and a slow response. Restore visibility, speed the response, and modernise detection with the team they have.

Your task. For each pain point below, name the requirement behind the concern, choose the product from the cheat-strip, then write how it helps and the outcome it delivers.

PAIN POINT 1 Case Study 3

> *When our security tools detect a compromised device, our manual response time is too slow. By the time we identify the user and manually isolate the port, the attacker has already moved through the network.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 2 Case Study 3

> *Our firewall environment has become too complex to manage manually.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 3 Case Study 3

> *Most of our traffic is encrypted, so we've lost visibility into a huge part of our environment.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 4 Case Study 3

> *We're worried traditional signature-based detection won't catch new or unknown attacks quickly enough.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

4 Case Study 4

The conversation. A flat data centre and fragmented Kubernetes networking invite east-west movement. Build segmentation Cedarline can adopt safely, with runtime visibility and application-dependency awareness so nothing breaks.

Your task. For each pain point below, name the requirement behind the concern, choose the product from the cheat-strip, then write how it helps and the outcome it delivers.

PAIN POINT 1 Case Study 4

> *Network policy helps, but we also need to know what's happening at runtime inside the workload. We need better detection of suspicious behavior and policy violations.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 2 Case Study 4

> *Our Kubernetes networking feels fragmented across clusters and environments. We're dealing with too many moving parts, inconsistent policies, and too much operational complexity.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 3 Case Study 4

> *Right now we're stitching together separate tools for networking, security, and observability in Kubernetes, and it's creating too much overhead.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

PAIN POINT 4 Case Study 4

> *We know we need microsegmentation, but we don't have enough visibility into east-west flows or application dependencies to create policy safely. We don't want to break the application.*

THE REQUIREMENT	
PRODUCT / PLATFORM	
HOW IT HELPS, AND THE OUTCOME	

Your Hybrid Mesh Firewall design

Four conversations, one story. Combine your case studies into a single HMF design and paste your team's topology below. Your facilitator collects this for **Subject Matter Expert (SME) review**: it is reviewed, not scored.

Paste your team's topology diagram here

Show how the four case studies connect across Cedarline's on-prem, data-centre and multicloud estate. Submitted for SME review.

Proctor evaluation

This page is completed by the proctor after your team submits the Workbook. Each case study is scored out of 100 against the Grading Guide, with feedback that supports the score, for an overall score out of 400.

Team ____________________ **Members (up to 8)** ____________________ **Date** ____________

CASE STUDY	SCORE / 100	FEEDBACK FOR THIS CASE STUDY
Case Study 1	______	
Case Study 2	______	
Case Study 3	______	
Case Study 4	______	
Overall total	______ **/ 400**	

Overall grade ____________________ (Best 300 to 400 · Better 200 to 299 · Good up to 199)

Topology diagram: SME review. The team's Hybrid Mesh Firewall diagram is submitted for Subject Matter Expert review and is not part of the score. Reviewed by SME: Yes ☐ No ☐

What the team did well

Where they can learn and grow